Enhancing Global Crisis Communication via Generative AI Chatbots: The Importance of Personalized Crisis Communication

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Abstract

Contemporary crises have highlighted the need for effective global crisis communication, however, this is often hindered by cognitive barriers that greatly limit public comprehension and cooperative engagement during a crisis. The consequences of crisis communication strategies that neglect these aspects of crises become especially apparent during high-stakes crises such as pandemics and climate crises. This thesis aims to explore how the newest developments in generative artificial intelligence (AI) chatbots show promising potential for enhancing crisis communication strategies of transnational organizations through personalized crisis messaging. Drawing from literature in crisis communication, cognitive psychology, and artificial intelligence, the paper examines how cognitive interferences – such as information overload, cognitive biases, perceived relevance, and cultural differences – to effective global crisis communication can be overcome by leveraging the emerging capabilities and applications of generative AI chatbots. More specifically, the discussion demonstrates how generative AI chatbots can complement and improve existing crisis communication methods by tailoring content, simulating empathy, and modifying message framing to individual users, ultimately increasing global public understanding and helpful participation. This thesis concludes by raising concerns regarding misinformation, misuse, and scalability, as well as other key limitations and ethical considerations for future research and practical applications. Through combining psychological theory with technological innovations, the study offers a human-computer interaction (HCI) centered framework for the beneficial integration of generative AI into global crisis communication strategies.

Keywords: generative AI, chatbots, crisis communication, personalized messaging, cognitive barriers, emotion, information processing, linguistic barriers

Introduction

Background on Global Crisis Management and Communication

Within the last decade, catastrophic global crises ranging from pandemics to climate change-induced natural disasters to armed conflicts have highlighted the fatal outcomes of ineffective global crisis management (GCM). Global crisis management[1] is defined as an amalgamation of four key factors –prevention, preparation, response, and revision– specifically "designed to combat crises and to lessen the actual damages inflicted by a crisis" (Coombs, 2015; Coombs & Laufer, 2018). GCM frameworks typically address multiple developmental stages of a crisis, an acknowledgement that is crucial to effectively target particular challenges in crisis management. Depending on the theoretical framework referenced, the number of stages may vary. According to Coombs et al. (2010), there are three main stages addressed during crisis management: a pre-crisis phase (prevention and preparation), a crisis phase (response), and a post-crisis phase (revision), whereas Reynold & Seeger's (2005) Crisis and Emergency Risk Communication Model (CERC) posits that there are 5 stages of a crisis (i.e., precrisis, initial event, maintenance, resolution, and evaluation) that must be addressed. While these stages guide large-scale institutional coordination to thoroughly address a crisis from before it begins to after the emergency event has passed, effective dissemination of crisis-related information is a critical aspect of every phase that ensures the public's safety. Thus, crisis communication functions as a central mechanism throughout global crisis management.

Crisis communication is "the collection, processing, and dissemination of information required to address a crisis" (Coombs et al., 2010), and typically involves "verbal, visual, and/or written interaction between the organization and its stakeholders" throughout all stages of a crisis(Fearn-Banks, 2002). In complex and high-stakes situations, the dissemination of timely, consistent, and transparent information is crucial for mitigating the effects of a global crisis (Becker, 2004; Branda et al, 2025). In addition, the initial messages conveyed by an organization during a crisis, specifically their receptibility by members of the public, are pivotal for effective crisis management (Benoit, 1997; Centers for Disease Control and Prevention, 2019; Coombs, 2022).

Several leading theoretical frameworks guide the discourse around global crisis communication. The Situational Crisis Communication Theory (SCCT), which posits that the optimal crisis response takes situational influences or contextual factors into account (Coombs, 2007, 2022), emphasizes the importance of assessing damages to an organization's reputation and public attributions of responsibility when formulating crisis communication strategies (Zhao et al., 2020). In contrast, the Discourse of Renewal (DOR) Theory (Ulmer et al., 2010), emphasizes reflective and growth-focused organizational learning from a crisis and suggests that effective crisis communication should incorporate these aspects of a crisis in a transparent and hopeful manner. All together, these theories accentuate the critical role of effective crisis communication in shaping crisis outcomes.

The Connection Between Cognition and Crisis Communication

According to Louise K. Comfort (2007), there are four processes – cognition, communication, coordination, and control – that have been identified as critical factors of effective crisis management. Comfort describes cognition, "the capacity to recognize the degree of emerging risk to which a community is exposed and to act on that information", as most critical of the four, as it is the trigger for other necessary processes for effective crisis management. Moreover, the proper response to and recovery from a crisis relies on understanding the strong bidirectional dependence between cognition and communication, which allows for the crucial flexibility for adapting to the dynamic conditions of a crisis (Comfort, 2007). The cognitive-based SCCT (Coombs, 2007, 2022), also highlights the importance of considering cognition when discussing crisis communication. The existing body of literature on crisis communication contains a considerable number of studies that have sought to demystify the relationship between cognition and communication (Balarezo et al., 2024; Coombs, 2022; Kim & Yang, 2009), yet the literature remains limited in establishing a clear relationship and the applications of it (Lu & Huang, 2018).

Limitations of Traditional Crisis Communication Strategies

The chaos that ensued from the coronavirus pandemic, a result of evidently inconsistent public service announcements, unclear health protocols, and slow coordination between federal and state governments, revealed the shockingly severe inadequacies of the current crisis communication strategies by transnational organizations (i.e., World Health Organization, United Nations) to the public. If the first step in an effective crisis response is addressing the physical and psychological safety of those who are impacted by a crisis (Coombs, 2007), it is important that crisis messages are directly relevant to the victims they are being communicated to – this highlights a key issue among current crisis communication strategies.

Due to the dynamic nature of a crisis and the people involved, organizations can not adhere to a universally applicable and uniform crisis response plan (Coombs & Holladay, 2009; Ulmer & Pyle, 2016; Reynolds & Seeger, 2005). This becomes especially relevant when developing global crisis communication methods, as the audience and environmental contexts become more diverse. The position of a transnational organization gives rise to a seemingly paradoxical situation where communications must balance breadth of reach and minimize misinformation and misinterpretation. A persistent challenge in global crisis communication is the consequences of generalized messaging; traditional approaches often employ generalized messages upon recognizing the onset of a crisis in hopes of broadening the reach of information to a wider audience. The "one-size-fits-all" methodology is a prominent point of criticism in literature concerning global public relations and communication, as it can exacerbate issues of misinformation and misinterpretation by overlooking the diverse ways individuals process and internalize information based on their unique characteristics and contexts (Hannes et al., 2024; Joseph, 2023; Overton et al., 2021). Since generalized crisis communication does not address these distinctions between people, it often prevents and reduces the exhibition of cooperative behavior during a crisis.

Current Applications of Artificial Intelligence in Crisis Management

An emerging area of research for improving crisis management is the use of artificial intelligence (AI) and machine learning (ML). Within the last decade, AI-driven systems have fueled a paradigm shift in crisis preparedness, response, and recovery efforts by emergency organizations from reactionary to anticipatory (Beduschi, 2022). For example, the United Nations Office for the Coordination for Humanitarian Affaris (OCHA) (2021) was able to proactively detect and assess natural disasters through AI-powered Geographic Information Systems (GIS) for mapping areas affected by cyclones in Mozambique, as well as predictive analytics forecasting floods in India and Bangladesh and famine in Somalia. Despite the proven and tangible benefits, applications of AI have been focused on using narrow AI and large-scale computation; with the newest developments in AI chatbots, the possible benefits have increased exponentially. The commonplaceness of human-AI interaction combined with the new generation of generative AI (GenAI) chatbots that are much more interactive and advanced than their pre-scripted predecessors presents a novel domain of research that could potentially save lives.

Research Gap and Objective

Although there is currently substantial research on crisis management and communication, the existing body of literature is unfortunately fragmented across disciplines (Auer et al., 2016), only focuses on case studies (Dhanesh & Sriramesh, 2016; Lyu, 2012; Nohrstedt, 2016), and concentrates on rebuilding strategies for profit and corporate businesses (Olsson, 2014). Additionally, despite acknowledgements of the importance of cognition in effective crisis communication and of the beneficial applications of artificial intelligence in crisis management, there is a notable gap in research combining the two areas, which may partially be due to the novelty of GenAI chatbots. This is indicative of an underexplored critical area of inquiry that may have significant practical implications.

As the world becomes increasingly interconnected, the need for globally applicable communication strategies is becoming undeniably necessary for preventing fatal crisis outcomes. By providing an interdisciplinary synthesis and analysis of literature in international crisis communication, cognitive psychology, and technological advancements in artificial intelligence, this paper aims to illustrate the promising potential of harnessing the abilities of GenAI chatbots to address areas of inadequacy in current international crisis communication strategies. The following sections will examine the underpinning mental processes of several cognitive barriers individuals face in international crisis communication, the newest capabilities of GenAI chatbots in terms of enhanced personalization, and discuss how a synergy of the two can be beneficial to incorporate in future crisis communication methods by humanitarian international organizations.

Literature Review

Cognitive Barriers in Crisis Communication

When individuals face a choice under uncertain circumstances, such as a crisis, they can experience subconscious errors in processing and understanding information they are presented with, "[separating] the human decision-maker from the field of rational choice theory" (Grinbaum, 2006). These disruptions in the mental processes that stray the reception and perception of external stimuli from a deliberate and systematic process, which accounts for the phenomenon of human failure to believe and act upon an announcement of a catastrophe (Dupuy & Grinbaum, 2005). While there are a multitude of cognitive barriers commonly associated with ineffective crisis communication (Lu & Huang, 2018; Perghel & Psychogios, 2013; Waring et al, 2019), this section will outline three particular cognitive challenges pertaining to both the initial reception and processing of crisis information and generalized messages in international crisis communication: the consequences of information overload, the perceived irrelevancy of crisis information, and linguistic and cultural barriers.

Mental Shortcuts During Crises

A common rule that organizations follow when faced with a crisis is to provide the public with clear and concise messaging. This is because presenting an immense quantity of information to people can severely undermine the effectiveness of crisis communication by causing information overload (Eppler & Mengis, 2004). During a high-stakes crisis, the inability to manage crisis information overload becomes a cognitive barrier. This can be attributed to the limitations humans have to their mental capacity combined with high-stress situations that require rapid processing of large influxes of information, which can have detrimental effects on cognitive functioning and psychological well-being (Al-Kumaim et al., 2021; Savolainen, 2015; Stanley, 2021). Specific consequences of information overload include poor decision-making due to difficulties in filtering and prioritizing information (Shahrzadi, 2024) and overwhelming the working memory capacity (Arnold, 2023).

According to Kahneman and Tversky (1984), there are two different systems of processing that individuals can go through: the automatic, quick, and intuitive System 1 or the deliberate, systematic, and effortful System 2. To make sense of uncertain yet complex situations, especially under time constraints, people adopt the heuristic System 1 processing (Kahneman & Tversky, 1984). Mental heuristics are essentially cognitive shortcuts that allow individuals to quickly make assessments and judgments without fully processing the given information (Rogers & Pearce, 2016; Slovic et al., 2004). Although this short-circuiting helps reduce cognitive load, engaging in fallible heuristic processing can induce systematic errors, or cognitive biases (Murata et al., 2015; Tversky & Kahneman, 1974). Like mental heuristics, people are highly likely to be influenced by cognitive biases under crises due to the uncertainty, time constraints, and high risk level (Comes, 2016; Phillips-Wren et al., 2019). Among maladaptive mental shortcuts that commonly emerge during crisis communication, there are two categories that are especially salient – those concerning informational influences, such as confirmation and anchoring bias, and those amplified by emotional states.

Confirmation Bias and Anchoring Bias. One of the most frequently identified cognitive biases in crisis communication literature (Brooks et al., 2020; Paulus et al., 2022; Tao, 2018), confirmation bias causes individuals to selectively interpret and search for information that aligns with their values while simultaneously dismiss information that opposes their beliefs (Nickerson, 1998; Oswald & Grosjean, 2004). This preference for information confirming their pre-existing assumptions and beliefs that confirmation biases cause is especially relevant to crisis communication due to its potential for creating a negative feedback loop that can be worsened if it occurs with the anchoring bias. The anchoring bias describes the tendency for decision-makers to rely on and be strongly influenced by the first piece of information they are presented with (Furnham & Boo, 2011; Lieder et al., 2018; Tversky & Kahneman, 1974). If given biased information during a crisis, confirmation bias will magnify the preference for more biased data to confirm their inaccurate preconceptions, perpetuating the spread of misinformation or disinformation (Paulus et al., 2022; Zhou & Shen, 2022).

Heuristics and Emotional States. Although not directly considered a cognitive barrier, the literature suggests that emotions play a critical role in rational reasoning (Leiserowitz, 2006) and can influence risk assessment by shifting an individual away from systematic information processing through affect heuristics (Finucane et al., 2000; Slovic et al., 2004). According to Lu & Huang's (2018) emotion-cognition dual-factor model, when people experience high or low emotional intensity states, they tend to follow an emotion-oriented pattern of processing or a cognition-oriented pattern of processing, respectively. Additionally, the long length of time that high emotional intensity states last (Sonnemans & Frijda, 1994) in conjunction with the impact intensity of an emotion has on decision-making (Damasio, 1994) presents a significant challenge for crisis communication. Research has found that negative and high emotional states common during a crisis such as fear, stress, anger, and anxiety (Centers for Disease Control and Prevention, 2019) have a particularly strong impact on an individual's susceptibility to framing effects (Lu & Huang, 2018), misinformation (Nabi, 2003), and attention to information (Kim & Cameron, 2011).

Attention and Perceived Relevance

The literature posits that perceived relevance or resonance of a crisis message may have a significant influence on an individual's willingness to process it. According to Petty & Cacioppo's (1981) elaboration likelihood model (ELM), people tend to process information in a more deliberate and systematic manner when it is perceived to be personally relevant to them. In comparison to generic communication content that is produced for a wide audience, the potential for mobilization in individuals increases when the content aligns with their beliefs or needs, possibly attributed to the gratification individuals feel when confirming their salient identities

(Wan, 2008). In line with this theoretical model, studies within crisis communication research have also found that message resonance is a mediating factor between the interpretation of the information and the subsequent behavioral outcome (Wan, 2008) and that attending to information is positively correlated with behavioral intentions (Kim, 2016). Within crisis contexts, this indicates that when messages communicated by emergency organizations are perceived to directly apply to the receiving individual or public, the likelihood of thorough information processing and taking action aligned with the message is higher. This difference in attention to crisis information depending on perceived relevance has also been supported by a study on perception of global risks by Weber (2006), which found that individuals perceive climate change as less personally relevant thus express less concern and supportive engagement. On the other hand, if a crisis message transmitted by an international organization contains instructions or information that directly contradict an individual's existing beliefs and values, it can cause cognitive dissonance (Wood & Miller, 2021), which can distort risk perception (Timar et al., 2014).

Linguistic and Cultural Differences

An important aspect of crisis communication that must be considered specifically in the context of international organizations such as the United Nations and the World Health Organization is that they are specifically trying to reach a diverse audience. Global diversity is extensive; in fact, linguists have found that there are 7,000 languages currently spoken globally (Eberhard, 2025), and anthropologists estimate that there are 4,000 distinct cultures is a severe underestimation of the total number of cultures present among the human population (Price, 1990). Studies have suggested that determinants of this extreme diversification in language and culture may be environmental influence (Birdsell, 1953; Foley & Lahr, 2011; Nettle, 1998), indicating that regional differences are a crucial aspect of reach to consider for global communication.

Language and cultural differences present a crucial challenge for international crisis communication in that they affect the initial accessibility and reception of a message. Regardless of the pertinence or urgency of processing crisis-related information, if it is in a language or utilizes cultural framing that is foreign to an individual, it would be incomprehensible and would not even reach evaluative and interpretative steps in information processing. Similar to the effect of perceived relevance aforementioned, when a crisis message is conveyed in a non-native or unfamiliar language, it can cause cognitive dissonance (Hatori et al., 2011), dampening the effectiveness of crisis communication. There is a similar diminishing effect when crisis communication does not consider cultural and political context (Xu & Huang, 2015). Language and cultural distinctions also play a key role in crisis communication.

For example, empirical studies have repeatedly proven the significance of appealing to emotion during communication. More specifically, expressing compassion for victims in crisis messaging can facilitate increased willingness to engage in support behavior (Augustine, 1995;

Coombs, 1999), and conveying sympathy and trust-building through transparency of organizational learning effectively engages the public and cultivates positive sentiments towards an organization (Zhao et al., 2020). Appealing to emotion would be completely ineffective if that compassion and sympathy can not be communicated past a linguistic barrier, which may greatly reduce the effectiveness of crisis communication by an organization.

Generative AI Chatbots for Personalized Communication

The widespread integration of artificial intelligence (AI) into personal and professional daily life, in conjunction with the exponential progression of GenAI over the past two years, denotes the proliferation of opportunities to transform the intelligent Human-Computer Interaction (HCI) landscape. Contrasting with earlier models where responses were rule-based and limited to scripts, newer GenAI chatbots are refined, essentially customizable agents that can engage in sophisticated conversation while considering and understanding contextual nuances.

Recognizing that the field of GenAI is continuously evolving, this section of the literature review will focus only on a few of the recent developments in GenAI chatbots that are pertinent to enhancing personalized communication. Concerning the core technologies powering the GenAI chatbots, the section will briefly cover several foundational technologies in connection to prominent recent augmentations (memory capacity, retrieval-augmented generation (RAG), and multimodality). The subsequent section will discuss the recently developed capabilities of the chatbots to: generate tailored content, sustain personalization through context-aware memory, and display emotional intelligence. Finally, recent applications of GenAI chatbots in diverse settings will be reviewed. These improvements represent significant progress in GenAI chatbots' capacity to process information and generate human-like responses that hold particular promise for facilitating more complex yet personable interactions with users.

Recent Developments in Infrastructure

Underlying the development of GenAI chatbots is a complex technological architecture involving several key technologies. Originally introduced by Vaswani et al. (2017), Large Language Models (LLMs), which are neural networks predominantly built using the Transformer architecture, are the core of GenAI chatbots. LLMs are rooted in self-attention mechanisms (Vaswani et al., 2017) trained using extensive data sets, which allow for the understanding of complex language and produce appropriate responses (Naveed et al., 2023; y Arcas, 2022). The advancements in LLMs are a product of the simultaneous progression in natural language processing (NLP) and machine learning (ML). Advancements in machine learning (ML) algorithms have allowed for deep learning, providing a multilayered training framework for LLMs to process and learn from complex datasets (Goodfellow, 2016). Natural language processing, utilized during the preprocessing stages of LLM training, provides the linguistic understanding GenAI chatbots require to interact with user queries using coherent dialogue (Adamopoulou & Moussiades, 2020; Raiaan et al., 2024). Recent advancements in AI have significantly enhanced the communicative capabilities of newer chatbot models. Some of the most significant are as follows.

Retrieval-Augmented Generation (RAG). The issue of responding with outdated or incorrect information in early chatbots was greatly attenuated through the introduction of retrieval-augmented generation (RAG), which enables AI systems to access and retrieve information from external knowledge databases (Lewis et al., 2020; Wu et al., 2024). This permits chatbots to be trained using external, domain-specific knowledge bases and respond to users with up to date information. The most recent development, Forward-Looking Active Retrieval augmented generation (FLARE), has significantly enhanced RAG by creating an iterative retrieval process, increasing the relevancy and reliability of responses generated by the chatbot (Jiang et al., 2023; Zhao et al., 2024) and refining natural language processing (Wu et al., 2024). This can be partially attributed to the RAG's mitigation of chatbot hallucinations, which presented a major challenge for factual accuracy in responses (Huang et al., 2025; Kang et al., 2024).

Multimodality. The recent addition of multimodality to LLMs – creating Large Multimodal Models (LMMs) – for AI chatbots enables them to generate content using multiple modalities (Naik et al., 2023; Ooi et al., 2023), a significant expansion from their previous limitation to text content generation. These more versatile models can process and generate creative content across different data types, such as images, videos, and audio, allowing for more nuanced, diverse, and comprehensive responses (Achiam et al., 2023; Naik et al., 2024).

Improved Memory. Memory plays a critical role in enhancing the capabilities of LLM-based agents and achieving computational universality (Hong & He, 2025; Schuurmans, 2023). Efficiency in memory mechanisms allows for the simulation of human behavior to extend past a single instance through the storing and retrieval of context-specific memories that relate to the user (Hong & He, 2025). Typically, memory refers to the retrieval of key-value pairs from a vector database. A system will selectively choose a contextually relevant memory aspect, which is a key, and subsequently extract the corresponding content, the value attached to the key (Tu et al., 2023). Those pieces of information fill a chatbot's context window, which is the maximum amount of textual information it can access and use at once (Brown et al., 2020). Whereas previous chatbot models had shorter context windows (Achiam et al., 2023), limiting the volume of potentially contextually relevant information the system could incorporate into a response, newer models now have improved memory through extended context windows.

Capabilities Specific to Personalization

Recent models, such as OpenAI's ChatGPT, Anthropic's Claude, Microsoft's Copilot, and Google's Gemini, have exhibited the behavioral outcomes of a refined infrastructure. Advancements in GenAI chatbots have equipped these systems with the ability to engage in personalized and sensitive interactions with users. Three key areas of recently enhanced capabilities are particularly important to highlight: creative content generation, sustained personalization, and emotion detection.

Sustained Personalization through Interaction History. The difficulty that earlier models had in offering consistent personalized social support that addresses diverse individual personalities and preferences (Tu et al., 2023) has been diminished by better contextual consistency facilitated by an extended context window. An extended context window allows chatbots to have an enriched understanding of a user by increasing the limit on the quantity of user input history that can be incorporated when generating a response. This indicates that models can utilize relevant information from even earlier interactions with a user to respond to a user's current query. Like a human's working memory (Graves et al., 2014), the extended context is similar to enabling longer memory retention in a chatbot. This means that chatbots can maintain contextual continuity across an increasing number of conversations with a user. In addition, multimodal systems significantly contribute to the personalization of GenAI chatbots by enabling an enriched understanding of a user's needs and behavior over time through a wider range of user inputs, such as auditory or visual (Wei et al., 2024). Collectively, these abilities help responses stay personalized for a given user past a single conversation, which is in essence, sustained personalized communication.

Generating Tailored Content for User Comprehension and Clarity. Although GenAI chatbots have previously been able to generate content relevant to the user, recent advancements have markedly improved the extent of content customization of responses. More specifically, the chatbots now have the tools to generate content that is tailored to the preferences and needs of a user. The integration of multimodality supports the cross-modal generation of creative responses in whichever form –text, images, video, or audio– most aligns with a user's needs and preferences (Bieniek et al., 2024; Singh & Singh, 2025). With improved RAG and extended context awareness, GenAI has also enhanced summarization capabilities (Chen et al., 2023; Zhao et al., 2024) that can quickly provide users with condensed yet accurate information in a more digestible form. Additionally, advancements in RAG strategies are equipping chatbots with the ability to execute efficient multilingual retrieval and real-time translation of retrieved information (Li et al., 2025; Ranaldi et al., 2025; Zhao et al., 2024), which enables the rapid adaptation of response content to user cultural and linguistic backgrounds, increasing user comprehension.

Emotion Detection. Although machines have been able to detect and respond to emotions expressed in text using sentiment analysis (Turney et al., 2003), they were based on rule-based systems that utilized manually specified lexicons and semantic patterns (Gupta, 2024). With the introduction of deep learning techniques to natural language processing, sentiment analysis has become increasingly sophisticated (Liu et al., 2012; Zhang et al., 2015), but earlier chatbot models such as ChaGPT struggled to empathize and provide emotional support (Zhao et al., 2023). However, the integration of sentiment analysis into GenAI chatbots, along with the recent addition of processing multiple modalities of data, has greatly enhanced their artificial emotional intelligence. Chatbots now can detect and analyze subtle emotional nuances (Gupta, 2024). Through multimodality, GenAI chatbots are able to process and interpret emotional cues from inputs that are not textual (Radford et al., 2021; Tewel, 2022). This includes identifying and measuring vocal, physical, and physiological signals such as vocal tones, facial expressions, body gestures, and neural signals (Geeta et al., 2024; Khare et al., 2024).

The capability to show human-like understanding through empathetic interactions is crucial in providing an anthropomorphic experience (Rajagopal, 2019), which can increase approachability and engagement with GenAI chatbots (Waytz et al., 2014), as well as increase trust in the novel technology (Prageet et al, 2025). Moreover, research indicates that trust in an organization is positively correlated with the likelihood of a person engaging in voluntary actions for the organization that surpass their own immediate needs (Gong & Park, 2023; Kim et al., 2021; Van Tonder et al., 2018).

Real-Life Applications

Beneficial applications of personalized communication by GenAI chatbots have been discussed and demonstrated across various domains; however, the most noteworthy are within education and healthcare. In educational settings, the content-generating capability of GenAI chatbots has been utilized in art-focused STEAM classes to foster creativity (Lee et al., 2024) and improve user comprehension of learning analytic dashboards by generating contextualized explanations for visualizations (Yan et al., 2024). Furthermore, deploying chatbots to personalize education has shown positive effects on cognitive functioning (Chauncey & McKenna, 2023; Klarin et al., 2024; Pergantis et al., 2025) and can aid people with learning disabilities in virtual learning spaces (Sáiz-Manzanares et al., 2023).

Recently, enhanced emotional awareness has proved to be a particularly promising and significant tool for improving healthcare. Healthcare professionals benefit from chatbots equipped with multimodal capabilities in areas such as documenting detailed medical histories, monitoring patient progress, overcoming language barriers, and identifying and analyzing images of medical issues (Meskó, 2023). Additionally, a study by Siddals et al. (2024) on the use of GenAI chatbots for mental health support found that the technology feels like an "emotional sanctuary" to individuals, and is comparable benefits-wise to human therapy.

Discussion

Adapting Crisis Communication to Individual Needs

The review of the literature revealed several areas of convergence between the cognitive barriers individuals face in crisis communication and the capabilities offered by recent advancements in GenAI chatbots. This section will synthesize relevant insights from the reviewed literature on the cognitive challenges associated with international crisis communication and the recent advancements in GenAI chatbot capabilities. As international humanitarian organizations face growing pressures to effectively communicate critical information to a diverse public, novel chatbot technologies provide innovative ways to bridge the gaps that currently exist in crisis communication strategies. GenAI chatbots offer a potential solution to the issues caused by generalized crisis messages by enabling interactive and personalized communication. Unlike traditional methods, these chatbots are capable of tailoring information to an individual user's qualities, including learning style, emotional needs, and situational contexts. This personalization may aid in mitigating the flaws in traditional crisis communication strategies by clarifying critical information in real-time, adjusting message tone and delivery to align with the user's emotional state and beliefs, and translating crisis information into local languages or idioms.

Addressing Comprehension of Information

One of the most promising advancements in the abilities of GenAI chatbots is their expanding capacity to tailor responses based on an individual's needs, preferences, and context. This capability directly addresses the result of a core cognitive challenge in crisis communication: inhibited crisis information comprehension. As reviewed in the literature, cognitive barriers such as information overload and low perceived relevance reduce message effectiveness during a crisis. Personalized communication by GenAI chatbots can help overcome these barriers by delivering information in a more digestible format.

GenAI chatbots with enriched understanding of users through improved memory capacities can quickly recall past interactions and adjust the structure, tone, and complexity of their responses to a query accordingly. In the context of crisis communication, this presents an opportunity for a more efficient approach to deliver urgent messages quickly while maintaining consistency in content, yet adapting the presentation of a message for individual preference or different groups of people. For instance, the same instructions for a public health protocol regarding disease spread prevention can be displayed in simple infographics for a visual learner while simultaneously being presented with detailed scientific explanations to an individual with a STEM background. Moreover, these chatbots can immediately recognize when someone is seeking clarification and follow a directive with iterative explanations in real-time that are increasingly tailored for the user's seamless comprehension.

Enhanced personalized communication by novel chatbot models also has implications for mitigating the cognitive effects of perceived irrelevance. During rapidly evolving crisis circumstances, people often experience impediments to processing information that conflicts with their values or is not framed in a manner that resonates with them. GenAI chatbots that can adjust the content of their responses to better align with a user's particular knowledge framework may aid in increasing mobilization and reducing internal conflicts about recommended actions. In addition, the use of GenAI chatbots to track individual, community-level, national, and even global engagement trends could rapidly inform broader communication strategies by international organizations, ensuring that the messages disseminated through different channels evolve continuously and in real-time with audience needs, without direct human interference. In short, the capability to tailor content on an individual basis represents a cognitively attuned yet scalable solution to the challenges concerning audience segmentation in global crisis management.

Furthermore, GenAI chatbots can serve as supportive, interactive fact-checking tools that address the inherent disadvantages of a human-led emergency response. During a crisis, human resources are often overwhelmed and spread thin due to staffing shortages or information bottlenecks, resulting in the inability to sufficiently address urgent public needs and questions. In contrast, chatbots offer continuous 24/7 support that can efficiently handle large volumes of inquiries. When deployed effectively, these chatbots could help address issues such as misinformation, confusion, and decision paralysis at the individual level. By providing users with immediate responses and clarifications, transnational organizations can reduce the ambiguity of a crisis and subdue public agitation, conceivably increasing public cooperation.

Expressing Emotional Sensitivity

Addressing the cognitive challenges in crisis communication is not solely a matter of delivering accurate, concise, and quick information – it also involves proper regulation of the arising emotions in affected populations. The significant influence that emotional states such as fear, anxiety, and anger have on information processing indicates the need for empathetic communication by international organizations during a crisis. Additionally, as demonstrated in the literature, intensely negative emotions can lead to a harmful reliance on mental heuristics and inhibit the attentive and thoughtful processing of information. Thus, the enhanced capacity of GenAI chatbots to display emotional awareness and sensitivity could substantially improve their effectiveness as tools for crisis communication.

Newer chatbot models have made significant strides in natural language processing and understanding, which now enables them to detect subtle emotional cues and nuances of users and respond with the appropriate levels of empathy. During a global crisis, the deployment of chatbots that can aid in public members' emotional regulation by offering personalized comfort, validating an individual's feelings, and using emotionally sensitive language could significantly improve the perceived relevancy and/or reception of crisis messages involving evacuations or quarantines.

In practical applications, emotionally sensitive responses could mitigate the sense of alienation often experienced during global crises, especially among more vulnerable or marginalized populations. For instance, a GenAI chatbot providing crisis information to a refugee might humanize its tone to convey compassion and address panicked frustrations without deviating from the message's core intent. It is important to note that functions such as those of artificial emotional intelligence are not replacements for human empathy, but only a supplementary layer that can be scaled rapidly across platforms to reach a wider audience.

When integrated with the directions of international humanitarian organizations and trained by psychological principles, these enhanced emotionally intelligent chatbots could play a pivotal role in maintaining social order during high-stress events.

Cultural and Linguistic Adaptation

Another crucial aspect of the ongoing improvements of GenAI chatbots is their capacity to adapt to cultural and linguistic contextual nuances. Global crises such as climate change are not limited to a singular nation, thus, international organizations must strive to reach diverse populations while retaining the integrity of a crisis message across them. More specifically, organizations must be able to provide real-time updates or safety instructions in multiple languages without compromising the quality of the message, yet in a way that affirms an individual's identity. As discussed in the literature review, cultural and linguistic barriers have been frequently identified in crisis communication research as a major obstacle to effectiveness; generalized crisis communications that neglect culture-specific norms, values, or nuanced language meanings when framing a message may be overlooked or outright rejected. Fortunately, GenAI chatbots offer a potentially transformative solution.

The improvements in multilingual language models and enhanced contextual awareness enable chatbots to deliver information in a user's native language and frame message content in a culturally salient manner. Capabilities such as these are essential for increasing the perceived relevance, which is a determining factor of whether information is processed deliberately or loosely, of crisis messages. By using linguistic nuances and conforming to an individual's cultural context – such as by avoiding using cultural taboos – GenAI chatbots can significantly enhance the effectiveness of international crisis communications.

Illustrative Case Studies

A recent experiment conducted by Liu, Huang, and Zhao (2025) exploring the impact of culturally tailored chatbots on encouraging information-seeking and mobilization during a disaster underscores the advantages of personal context-aware messaging in a crisis situation. In this study, a sample of 346 African American and Hispanic residents from Texas and Florida were provided culturally tailored hurricane preparedness information through GPT-4 configurations (Liu et al., 2025). Study findings indicated that chatbots that mirror a user's specific cultural background are perceived as more credible and promote information sharing and preparedness behavior, potentially as a result of homophily and increased source credibility (Liu et al., 2025). Notably, the effectiveness of culturally tailored chatbots varied across ethnic groups (Liu et al., 2025), demonstrating the importance of nuanced communication strategies when striving for global reach. This empirical study strongly supports the benefits of leveraging GenAI chatbots to address the cognitive gaps in crisis communication and affirms the practicality of integrating novel chatbot models into global crisis communication frameworks.

In addition, a pertinent real-world application example of personalized communication through generative AI is India's 2024 election. According to a national census, India has 22 officially recognized national languages and over 19,500 regional dialects (Press Trust of India, 2018), thus, navigating the broadcast of political campaigns presents a significant challenge. Rather than the customary door-to-door campaigning political candidates engaged in to reach voters individually, advancements in AI and the widespread internet use presented a much more economical solution (Dhanuraj et al., 2024). To expand voter outreach to more remote communities, Indian political parties during the 2024 national election utilized AI to deliver candidates' messages in a more personalized manner (Christopher & Bansal, 2024). Through generative AI, a candidate's voice would be used to generate videos and calls that could be sent to individuals' phones. These deepfakes of the political candidate would directly address the voter with tailored and personal context, including the voter's name and references to political issues they have shown interest in (Dhanuraj et al., 2024). Although the use of generative AI facilitated broader outreach, it remains the subject of considerable controversy during the election due to troubling concerns regarding the misuse of deepfakes for the dissemination of propaganda and misinformation (Shukla & Tripathi, 2024). Nevertheless, the application of GenAI in India's 2024 election provides invaluable insight into the considerable potential of integrating such technologies into global crisis communication efforts.

Limitations

While recent advancements in GenAI chatbots offer a compelling opportunity to enhance global crisis communication, there are several limitations that must be acknowledged to provide a comprehensive picture of the feasibility of such an integration. First, there is a substantial lack of empirical research and evidence within this area of study due to its emerging nature. Much of the current literature on the recent advancements in and applications of GenAI is awaiting review or is purely speculative, leaving crucial questions concerning the feasible implementation and effectiveness of the new technology largely unanswered. Over-reliance on unfounded assumptions can undermine public trust in organizations deploying these methods, as well as weaken the credibility of the field of research itself. Secondly, although these AI systems are inherently scalable, deployment at a global scale poses a logistically complex challenge, as it likely to be inhibited by systemic inequalities in access to novel technologies and differences in digital infrastructure across nations. Furthermore, the incorporation of GenAI into crisis communication strategies raises critical ethical issues that must be considered. National cybersecurity, as well as infringements on personal liberties, including data protection and privacy, have become notorious concerns. Additionally, the computational power necessary to train AI models and manage hardware demands requires immense quantities of electricity and water, which can not only increase carbon dioxide emissions but disrupt ecosystems (Bashir et al., 2024).

Moreover, there are still pertinent limitations of AI chatbots and novel models – particularly with hallucinations, susceptibility to manipulation by human developers, and training using inevitably biased data. Regardless of improvements in reasoning skills and agency in artificial intelligence systems, close human supervision is still necessary to ensure regulatory compliance and to verify the accuracy and appropriateness of chatbot responses (Naik et al., 2023, 2024). AI technology is fallible, and malfunctions, disinformation, and propagandized training, especially made apparent during a high-stakes crisis, pose a significant threat to the integrity and reputation of global institutions. These limitations stress the pressing need for robust interdisciplinary research if GenAI chatbots are to play a meaningful role in global crisis communication efforts that they are capable of fulfilling.

Conclusion and Future Directions

Multifaceted issues require interdisciplinary research and solutions. As global crises become increasingly unpredictable and complex, this thesis argues that improving global crisis communication must take a holistic approach. It is imperative for international humanitarian organizations to not only consider the cognitive mechanisms that guide human information processing, reasoning, and behavior, but to proactively incorporate novel technologies as a tool to support those mental processes. The capacity of GenAI chatbots to frame information in ways that align with a user could significantly influence engagement and preparedness outcomes of a global crisis.

As with international organizations, whose crisis responses hinge on a rapid yet homogenous perception of the crisis itself (Hirschmann, 2023), developers of crisis communication strategies must consider how people internally process and interpret crisis events. It is essential for further research to be done on the cognitive dimensions of global crisis communication, as cognition is the root of human existence. Additionally, future research should explore how such AI systems can be effectively deployed through robust empirical studies conducted in controlled environments. The rapid advancements in generative AI technology and the realization of agentic AI are reshaping the technological landscape at unprecedented speeds, and this calls for research focused on utilizing updated information to look into the ethical and optimal method of use across varied contexts.

The goal of crisis communication should not be compliance, but cooperation: compliance is forced and temporary, whereas cooperation is fueled by understanding. Transnational organizations must foster an intrinsic motivation for social cooperation at times of crisis. At a time where geopolitical fragmentation is growing and societal cohesion is deteriorating, the protection of human life – and humanity itself – should transcend divisions and be a unifying principle. Generative AI chatbots, with their ability to personalize communication without sacrificing scalability, may serve as a valuable tool in global crisis communication to safeguard human life during times of imminent danger.

References

Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., ... & McGrew, B. (2023). *GPT-4 technical report* [Preprint]. *arXiv*. https://arxiv.org/abs/2303.08774

Adamopoulou, E., & Moussiades, L. (2020). An overview of chatbot technology. In I. Maglogiannis, L. Iliadis, & E. Pimenidis (Eds.), *Artificial Intelligence Applications and Innovations* (Vol. 584, pp. 373–383). Springer. https://doi.org/10.1007/978-3-030-49186-4_31

Al-Kumaim, N. H., Hassan, S. H., Shabbir, M. S., Almazroi, A. A., & Al-Rejal, H. M. A. (2021). Exploring the inescapable suffering among postgraduate researchers: Information overload perceptions and implications for future research. *International Journal of Information and Communication Technology Education*, *17*(1), 19–41.

Arnold, M., Goldschmitt, M., & Rigotti, T. (2023). Dealing with information overload: A comprehensive review. *Frontiers in Psychology*, *14*, 1122200. https://doi.org/10.3389/fpsyg.2023.1122200

Auer, C., Schwarz, A., & Seeger, M. W. (2016). Communication – Conclusions for an integrative approach to international crisis communication research. In A. Schwarz, M. W. Seeger, & C. Auer (Eds.), *The handbook of international crisis communication research* (pp. 66–71). Wiley-Blackwell. https://doi.org/10.1002/9781118516812.ch7

Augustine, N. R. (1995). Managing the crisis you tried to prevent. *Harvard Business Review*, *73*(6), 147–158.

Balarezo, J. D., Foss, N. J., & Nielsen, B. B. (2024). Organizational learning: Understanding cognitive barriers and what organizations can do about them. *Management Learning*, *55*(5), 741–768. https://doi.org/10.1177/13505076231210635

Bashir, N., Donti, P., Cuff, J., Sroka, S., Ilic, M., Sze, V., ... & Olivetti, E. (2024). *The climate and sustainability implications of generative AI* [Preprint]. *arXiv*.

Beduschi, A. (2022). Harnessing the potential of artificial intelligence for humanitarian action: Opportunities and risks. *International Review of the Red Cross*, *104*(919), 1149–1169. https://doi.org/10.1017/S1816383122000261

Benoit, W. L. (1997). Image repair discourse and crisis communication. *Public Relations Review*, *23*(2), 177–186. https://doi.org/10.1016/S0363-8111(97)90023-0

Bieniek, J., Rahouti, M., & Verma, D. C. (2024). Generative AI in multimodal user interfaces: Trends, challenges, and cross-platform adaptability. *arXiv*. https://arxiv.org/abs/2411.10234

Birdsell, J. B. (1953). Some environmental and cultural factors influencing the structuring of Australian Aboriginal populations. *The American Naturalist*, *87*(834), 171–207.

Branda, F., Stella, M., Ceccarelli, C., Cabitza, F., Ceccarelli, G., Maruotti, A., Ciccozzi, M., & Scarpa, F. (2025). The role of AI-based chatbots in public health emergencies: A narrative review. *Future Internet*, *17*(4), 145. https://doi.org/10.3390/fi17040145

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, *33*, 1877–1901. https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf

Centers for Disease Control and Prevention [CDC]. (2019). *CERC: Psychology of a crisis*. https://www.cdc.gov/cerc/media/pdfs/CERC_Psychology_of_a_Crisis.pdf

Chauncey, S. A., & McKenna, H. P. (2023). A framework and exemplars for ethical and responsible use of AI chatbot technology to support teaching and learning. *Computers and Education: Artificial Intelligence*, *5*, 100182. https://doi.org/10.1016/j.caeai.2023.100182

Chen, S., Wong, S., Chen, L., & Tian, Y. (2023). Extending context window of large language models via positional interpolation. *arXiv*. https://arxiv.org/abs/2306.15595

Christopher, N., & Bansal, V. (2024). Indian voters are being bombarded with millions of deepfakes. Political candidates approve. *WIRED*. https://www.wired.com/story/indian-elections-ai-deepfakes/

Comes, T. (2016, March). Cognitive biases in humanitarian sensemaking and decision-making: Lessons from field research. In *2016 IEEE International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA)* (pp. 56–62). IEEE. https://doi.org/10.1109/CogSIMA.2016.7497792

Comfort, L. K. (2007). Crisis management in hindsight: Cognition, communication, coordination, and control. *Public Administration Review*, *67*, 189–197. https://www.jstor.org/stable/4624696

Coombs, W. T. (1999). Information and compassion in crisis responses: A test of their effects. *Journal of Public Relations Research*, *11*(2), 125–142. https://doi.org/10.1207/s1532754xjprr1102_02

Coombs, W. T. (2015). *Ongoing crisis communication: Planning, managing, and responding* (4th ed.). Sage Publications.

Coombs, W. T. (2022). Situational crisis communication theory (SCCT): Refining and clarifying a cognitive-based theory of crisis communication. In T. Coombs & S. J. Holladay (Eds.), *The handbook of crisis communication* (pp. 193–204). John Wiley & Sons. https://doi.org/10.1002/9781119678953.ch14

Coombs, W. T., & Holladay, S. J. (2009). Further explorations of post-crisis communication: Effects of media and response strategies on perceptions and intentions. *Public Relations Review*, *35*(1), 1–6. https://doi.org/10.1016/j.pubrev.2008.09.011

Coombs, W. T., & Laufer, D. (2018). Global crisis management – Current research and future directions. *Journal of International Management*, *24*(3), 199–203. https://doi.org/10.1016/j.intman.2017.12.003

Coombs, W. T., Holladay, S. J., & Wiley, J. (Eds.). (2010). *The handbook of crisis communication*. Wiley-Blackwell.

Damasio, A. R. (2006). *Descartes' error: Emotion, reason, and the human brain*. Random House.

Dhanesh, G. S., & Sriramesh, K. (2016). Risk and crisis communication research in India. In A. Schwarz, M. W. Seeger, & C. Auer (Eds.), *The handbook of international crisis communication research* (pp. 302–312). John Wiley & Sons. https://doi.org/10.1002/9781118516812.ch28

Dhanuraj, D., Harilal, S., & Solomon, N. (2024). *Generative AI and its influence on India’s 2024 elections*. Centre for Public Policy Research.

Dupuy, J. P., & Grinbaum, A. (2005). Living with uncertainty: From the precautionary principle to the methodology of ongoing normative assessment. *Comptes Rendus Geoscience*, *337*(4), 457–474. https://doi.org/10.1016/j.crte.2004.10.015

Eberhard, D. M., Simons, G. F., & Fennig, C. D. (Eds.). (2025). *Ethnologue: Languages of the world* (28th ed.). SIL International.

Eppler, M. J., & Mengis, J. (2008). The concept of information overload: A review of literature from organization science, accounting, marketing, MIS, and related disciplines. In *Kommunikationsmanagement im Wandel: Beiträge aus 10 Jahren = mcminstitute* (pp. 271–305).

Fearn-Banks, K. (2002). *Crisis communications* (2nd ed.). Lawrence Erlbaum Associates.

Finucane, M. L., Alhakami, A., Slovic, P., & Johnson, S. M. (2000). The affect heuristic in judgments of risks and benefits. *Journal of Behavioral Decision Making*, *13*(1), 1–17. https://doi.org/10.1002/(SICI)1099-0771(200001/03)13:1<1::AID-BDM333>3.0.CO;2-S

Foley, R. A., & Lahr, M. M. (2011). The evolution of the diversity of cultures. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *366*(1567), 1080–1089. https://doi.org/10.1098/rstb.2010.0370

Geetha, A. V., Mala, T., Priyanka, D., & Uma, E. (2024). Multimodal emotion recognition with deep learning: Advancements, challenges, and future directions. *Information Fusion*, *105*, 102218. https://doi.org/10.1016/j.inffus.2024.102218

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Graves, A., Wayne, G., & Danihelka, I. (2014). Neural Turing machines. *arXiv*. https://arxiv.org/abs/1410.5401

Grinbaum, A. (2006). Cognitive barriers in perception of nanotechnology. *Journal of Law, Medicine & Ethics*, *34*(4), 689–694. https://doi.org/10.1111/j.1748-720X.2006.00189.x

Gupta, S., Ranjan, R., & Singh, S. N. (2024). Comprehensive study on sentiment analysis: From rule-based to modern LLM-based system. *arXiv*. https://arxiv.org/abs/2409.09989

Hannes, K., Thyssen, P., Bengough, T., Dawson, S., Paque, K., Talboom, S., Tuand, K., Vandendriessche, T., Wopereis, D., & Vandamme, M. (2024). Inclusive crisis communication in a pandemic context: A rapid review. *International Journal of Environmental Research and Public Health*, *21*(9), 1216. https://doi.org/10.3390/ijerph21091216

Hatori, T., Kobayashi, K., & Jeong, H. (2011). Public reviews and trust formation in disaster risk management. *Journal of Natural Disaster Science*, *32*(2), 89–97.

Hirschmann, G. (2023). Crisis management in international organisations: The League of Nations' response to early challenges. *Cambridge Review of International Affairs*, *37*(5), 573–591. https://doi.org/10.1080/09557571.2023.2271984

Hong, C., & He, Q. (2025). Enhancing memory retrieval in generative agents through LLM-trained cross attention networks. *Frontiers in Psychology*, *16*, 1591618. https://doi.org/10.3389/fpsyg.2025.1591618

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., ... & Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, *43*(2), 1–55. https://doi.org/10.1145/3649024

Jiang, Z., Xu, F. F., Gao, L., Sun, Z., Liu, Q., Dwivedi-Yu, J., ... & Neubig, G. (2023, December). Active retrieval augmented generation. In *Proceedings of the 2023*

Conference on Empirical Methods in Natural Language Processing (pp. 7969–7992). Association for Computational Linguistics.

Joseph, C. K. (2023). Cultural sensitivity in global public relations campaigns. *American Journal of Public Relations*, *2*(1), 20–27. https://doi.org/10.47672/ajpr.1629

Kahneman, D., & Tversky, A. (1984). Choices, values, and frames. *American Psychologist*, *39*(4), 341–350. https://doi.org/10.1037/0003-066X.39.4.341

Kang, M., Gürel, N. M., Yu, N., Song, D., & Li, B. (2024). C-RAG: Certified generation risks for retrieval-augmented language models. *arXiv*. https://arxiv.org/abs/2402.03181

Khare, S. K., Blanes-Vidal, V., Nadimi, E. S., & Acharya, U. R. (2024). Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations. *Information Fusion*, *102*, 102019. https://doi.org/10.1016/j.inffus.2023.102019

Kim, H. J., & Cameron, G. T. (2011). Emotions matter in crisis. *Communication Research*, *38*(6), 826–855. https://doi.org/10.1177/0093650210385813

Kim, H. K., & Yang, S. U. (2009). Cognitive processing of crisis communication: Effects of CSR and crisis response strategies on stakeholder perceptions of a racial crisis dynamics. *Public Relations Journal*, *3*(1), 1–39.

Kim, Y. (2016). Understanding publics' perception and behaviors in crisis communication: Effects of crisis news framing and publics' acquisition, selection, and transmission of information in crisis situations. *Journal of Public Relations Research*, *28*(1), 35–50. https://doi.org/10.1080/1062726X.2016.1145733

Klarin, J., Hoff, E. V., Larsson, A., & Daukantaitė, D. (2024). Adolescents' use and perceived usefulness of generative AI for schoolwork: Exploring their relationships with executive functioning and academic achievement. *Frontiers in Artificial Intelligence*, *7*, 1415782. https://doi.org/10.3389/frai.2024.1415782

Lee, U., Han, A., Lee, J., Lee, E., Kim, J., Kim, H., & Lim, C. (2024). Prompt Aloud!: Incorporating image-generative AI into STEAM class with learning analytics using prompt data. *Education and Information Technologies*, *29*(8), 9575–9605. https://doi.org/10.1007/s10639-024-12323-1

Leiserowitz, A. (2006). Climate change risk perception and policy preferences: The role of affect, imagery, and values. *Climatic Change*, *77*, 45–72. https://doi.org/10.1007/s10584-006-9059-9

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, *33*, 9459–9474.

Li, S., Stenzel, L., Eickhoff, C., & Bahrainian, S. A. (2025). Enhancing retrieval-augmented generation: A study of best practices. *arXiv*. https://arxiv.org/abs/2501.07391

Liu, B. (2022). *Sentiment analysis and opinion mining*. Springer Nature. https://doi.org/10.1007/978-3-031-16173-3

Liu, Y., Kauttonen, J., Zhao, B., Li, X., & Peng, W. (2024). Editorial: Towards emotion AI to next generation healthcare and education. *Frontiers in Psychology*, *15*, 1533053. https://doi.org/10.3389/fpsyg.2024.1533053

Liu, W., Huang, Y., & Zhao, X. (2025). Exploring the contingent effect of cultural tailoring of GenAI chatbots on multiethnic members' disaster information seeking and preparation attitude: A cross-group comparison. *SSRN*. https://doi.org/10.2139/ssrn.5208105

Lu, Y., & Huang, Y.-H. C. (2018). Getting emotional: An emotion-cognition dual-factor model of crisis communication. *Public Relations Review*, *44*(1), 98–107. https://doi.org/10.1016/j.pubrev.2017.09.007

Lyu, J. C. (2012). A comparative study of crisis communication strategies between Mainland China and Taiwan: The melamine-tainted milk powder crisis in the Chinese context. *Public Relations Review*, *38*(5), 779–791. https://doi.org/10.1016/j.pubrev.2012.07.003

Meskó, B. (2023). The impact of multimodal large language models on health care's future. *Journal of Medical Internet Research*, *25*, e52865. https://doi.org/10.2196/52865

Murata, A., Nakamura, T., & Karwowski, W. (2015). Influence of cognitive biases in distorting decision making and leading to critical unfavorable incidents. *Safety*, *1*(1), 44–58. https://doi.org/10.3390/safety1010044

Nabi, R. L. (2003). Exploring the framing effects of emotion: Do discrete emotions differentially influence information accessibility, information seeking, and policy preference? *Communication Research*, *30*(2), 224–247. https://doi.org/10.1177/0093650202250881

Naik, D., Naik, I., & Naik, N. (2024, July). Applications of AI chatbots based on generative AI, large language models and large multimodal models. In *The International Conference on Computing, Communication, Cybersecurity & AI* (pp. 668–690). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-53674-6_50

Naik, I., Naik, D., & Naik, N. (2023, September). Artificial intelligence (AI) applications in chemistry. In *UK Workshop on Computational Intelligence* (pp. 545–557). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-48222-7_45

Naveed, H., Khan, A. U., Qiu, S., Saqib, M., Anwar, S., Usman, M., ... & Mian, A. (2023). A comprehensive overview of large language models. *arXiv*. https://arxiv.org/abs/2307.06435

Nettle, D. (1998). Explaining global patterns of language diversity. *Journal of Anthropological Archaeology*, *17*(4), 354–374. https://doi.org/10.1006/jaar.1998.0328

Nohrstedt, S. A. (2016). The role of the media in the discursive construction of wars. In A. Schwarz, M. W. Seeger, & C. Auer (Eds.), *The handbook of international crisis communication research* (pp. 133–144). John Wiley & Sons. https://doi.org/10.1002/9781118516812.ch13

Olsson, E. K. (2014). Crisis communication in public organisations: Dimensions of crisis communication revisited. *Journal of Contingencies and Crisis Management*, *22*(2), 113–125. https://doi.org/10.1111/1468-5973.12049

Ooi, K. B., Tan, G. W. H., Al-Emran, M., Al-Sharafi, M. A., Capatina, A., Chakraborty, A., … Wong, L. W. (2025). The Potential of Generative Artificial Intelligence Across Disciplines: Perspectives and Future Directions. *Journal of Computer Information Systems*, *65*(1), 76–107. https://doi.org/10.1080/08874417.2023.2261010

Overton, D., Ramkeesoon, S. A., Kirkpatrick, K., Byron, A., & Pak, E. S. (Eds.). (2021). *Lessons from COVID-19 on executing communications and engagement at the community level during a health crisis*. National Academies Press. https://doi.org/10.17226/26340

Pergantis, P., Bamicha, V., Skianis, C., & Drigas, A. (2025). AI chatbots and cognitive control: Enhancing executive functions through chatbot interactions: A systematic review. *Brain Sciences*, *15*(1), 47. https://doi.org/10.3390/brainsci15010047

Perghel, R., & Psychogios, A. G. (2013). Making sense of crisis: Cognitive barriers of learning in critical situations. *Management Dynamics in the Knowledge Economy*, *1*(2), 179–194.

Petty, R. E., & Cacioppo, J. T. (1981). Epilog: A general framework for understanding attitude change processes. In R. E. Petty & J. T. Cacioppo (Eds.), *Attitudes and persuasion: Classic and contemporary approaches* (pp. 255–269). William C. Brown.

Phillips-Wren, G., Power, D. J., & Mora, M. (2019). Cognitive bias, decision styles, and risk attitudes in decision making and DSS. *Journal of Decision Systems*, *28*(2), 63–66. https://doi.org/10.1080/12460125.2019.1646509

Press Trust of India. (2018, July 2). More than 19,500 mother tongues spoken in India: Census. *Indian Express*. https://indianexpress.com/article/india/more-than-19500-mother-tongues-spoken-in-india-census-5241056/

Price, D. H. (1990). *Atlas of world cultures: A geographical guide to ethnographic literature*. Sage.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... & Sutskever, I. (2021, July). Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning* (pp. 8748–8763). PMLR.

Raiaan, M. A. K., Mukta, M. S. H., Fatema, K., Fahad, N. M., Sakib, S., Mim, M. M. J., ... & Azam, S. (2024). A review on large language models: Architectures, applications, taxonomies, open issues and challenges. *IEEE Access*, *12*, 26839–26874. https://doi.org/10.1109/ACCESS.2024.3381435

Ranaldi, L., Haddow, B., & Birch, A. (2025). Multilingual retrieval-augmented generation for knowledge-intensive task. *arXiv*. https://arxiv.org/abs/2504.03616

Reynolds, B., & Seeger, M. W. (2005). Crisis and emergency risk communication as an integrative model. *Journal of Health Communication*, *10*(1), 43–55. https://doi.org/10.1080/10810730590904571

Rogers, M. B., & Pearce, J. M. (2016). The psychology of crisis communication. In A. Schwarz, M. W. Seeger, & C. Auer (Eds.), *The handbook of international crisis communication research* (pp. 34–44). John Wiley & Sons. https://doi.org/10.1002/9781118516812.ch4

Sáiz-Manzanares, M. C., Marticorena-Sánchez, R., Martín-Antón, L. J., Díez, I. G., & Almeida, L. (2023). Perceived satisfaction of university students with the use of chatbots as a tool for self-regulated learning. *Heliyon*, *9*(1), e12644. https://doi.org/10.1016/j.heliyon.2022.e12644

Savolainen, R. (2015). Cognitive barriers to information seeking: A conceptual analysis. *Journal of Information Science*, *41*(5), 614–623. https://doi.org/10.1177/0165551515587850

Schuurmans, D. (2023). Memory augmented large language models are computationally universal. *arXiv*. https://arxiv.org/abs/2301.04589

Shahrzadi, L., Mansouri, A., Alavi, M., & Shabani, A. (2024). Causes, consequences, and strategies to deal with information overload: A scoping review. *International Journal of Information Management Data Insights*, *4*(2), 100261. https://doi.org/10.1016/j.jjimei.2024.100261

Shukla, A. K., & Tripathi, S. (2024). AI-generated misinformation in the election year 2024: Measures of European Union. *Frontiers in Political Science*, *6*, 1451601. https://doi.org/10.3389/fpos.2024.1451601

Siddals, S., Torous, J., & Coxon, A. (2024). "It happened to be the perfect thing": Experiences of generative AI chatbots for mental health. *NPJ Mental Health Research*, *3*(1), 1–9. https://doi.org/10.1038/s44184-024-00097-4

Singh, A., & Singh, K. K. (2025). Future directions in multimodal generative AI. In A. Singh & K. K. Singh (Eds.), *Multimodal generative AI* (pp. 335–350). Springer. https://doi.org/10.1007/978-981-96-2355-6_16

Slovic, P., Finucane, M. L., Peters, E., & MacGregor, D. G. (2004). Risk as analysis and risk as feelings: Some thoughts about affect, reason, risk, and rationality. *Risk Analysis*, *24*(2), 311–322. https://doi.org/10.1111/j.0272-4332.2004.00433.x

Sonnemans, J., & Frijda, N. H. (1994). The structure of subjective emotional intensity. *Cognition and Emotion*, *8*(4), 329–350. https://doi.org/10.1080/02699939408408945

Stanley, O. E. (2021). Information overload: Causes, symptoms, consequences and solutions. *Asian Journal of Information Science and Technology*, *11*(2), 1–6.

Tewel, Y., Shalev, Y., Schwartz, I., & Wolf, L. (2022). Zerocap: Zero-shot image-to-text generation for visual-semantic arithmetic. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 17918–17928).

Timar, L., Grimes, A., & Fabling, R. (2014). That sinking feeling: The changing price of disaster risk following an earthquake. *SSRN*. https://doi.org/10.2139/ssrn.2523204

Tu, Q., Chen, C., Li, J., Li, Y., Shang, S., Zhao, D., ... & Yan, R. (2023). CharacterChat: Learning towards conversational AI with personalized social support. *arXiv*. https://arxiv.org/abs/2308.10278

Turney, P. D., & Littman, M. L. (2003). Measuring praise and criticism: Inference of semantic orientation from association. *ACM Transactions on Information Systems*, *21*(4), 315–346. https://doi.org/10.1145/944012.944013

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, *185*(4157), 1124–1131. https://doi.org/10.1126/science.185.4157.1124

Ulmer, R. R., & Pyle, A. S. (2016). International organizational crisis communication. In A. Schwarz, M. W. Seeger, & C. Auer (Eds.), *The handbook of international crisis communication research* (pp. 108–118). John Wiley & Sons. https://doi.org/10.1002/9781118516812.ch11

Ulmer, R. R., Sellnow, T. L., & Seeger, M. W. (2010). *Effective crisis communication: Moving from crisis to opportunity*. Sage Publications.

United Nations Office for the Coordination of Humanitarian Affairs. (2021). *From digital promise to frontline practice: New and emerging technologies in humanitarian action*. https://reliefweb.int/report/world/digital-promise-frontline-practice-new-and-emerging-technologies-humanitarian-action

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.

Wan, H. (2008). Resonance as a mediating factor accounting for the message effect in tailored communication—Examining crisis communication in a tourism context. *Journal of Communication*, *58*(3), 472–489. https://doi.org/10.1111/j.1460-2466.2008.00395.x

Waring, S., Humann, M., & Dawson, N. (2019). Facilitators and barriers to effective information sharing during international disaster response. *Journal of Emergency Management*, *17*(6), 469–486. https://doi.org/10.5055/jem.2019.0445

Weber, E. U. (2006). Experience-based and description-based perceptions of long-term risk: Why global warming does not scare us (yet). *Climatic Change*, *77*(1), 103–120. https://doi.org/10.1007/s10584-006-9060-3

Wei, T., Jin, B., Li, R., Zeng, H., Wang, Z., Sun, J., ... & Tang, X. (2024). Towards unified multi-modal personalization: Large vision-language models for generative recommendation and beyond. *arXiv*. https://arxiv.org/abs/2403.10667

Wood, E., & Miller, S. K. (2021). Cognitive dissonance and disaster risk communication. *Journal of Emergency Management and Disaster Communications*, *2*(1), 39–56.

Wu, S., Xiong, Y., Cui, Y., Wu, H., Chen, C., Yuan, Y., ... & Xue, C. J. (2024). Retrieval-augmented generation for natural language processing: A survey. *arXiv*. https://arxiv.org/abs/2407.13193

Xu, K., & Huang, Y. H. (2015). Effects of message framing and source credibility on risk communication in China: A study of public reactions to the 2013 H7N9 outbreak. *Public Relations Review*, *41*(5), 721–732. https://doi.org/10.1016/j.pubrev.2015.09.003

y Arcas, B. A. (2022). Do large language models understand us? *Daedalus*, *151*(2), 183–197. https://doi.org/10.1162/daed_a_01925

Yan, L., Zhao, L., Echeverria, V., Jin, Y., Alfredo, R., Li, X., Gašević, D., & Martinez-Maldonado, R. (2024). VizChat: Enhancing learning analytics dashboards with contextualised explanations using multimodal generative AI chatbots. In A. M. Olney, I.-A. Chounta, Z. Liu, O. C. Santos, & I. I. Bittencourt (Eds.), *Artificial Intelligence in Education* (Vol. 14830, pp. 180–193). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-64299-9_13

Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level convolutional networks for text classification. *Advances in Neural Information Processing Systems*, *28*.

Zhao, P., Zhang, H., Yu, Q., Wang, Z., Geng, Y., Fu, F., ... & Cui, B. (2024). Retrieval-augmented generation for AI-generated content: A survey. *arXiv*. https://arxiv.org/abs/2402.19473

Zhao, W., Zhao, Y., Lu, X., Wang, S., Tong, Y., & Qin, B. (2023). Is ChatGPT equipped with emotional dialogue capabilities? *arXiv*. https://arxiv.org/abs/2304.09582

Zhao, X., Zhan, M., & Ma, L. (2020). How publics react to situational and renewing organizational responses across crises: Examining SCCT and DOR in social-mediated crises. *Public Relations Review*, *46*(4), 101944. https://doi.org/10.1016/j.pubrev.2020.101944